

Alfredo Mejia

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Education

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| • Computer Science, M.S. | The University of Texas at Austin | May 2024 |
| • Software Engineering, B.S. | The University of Texas at Dallas | <i>Summa Cum Laude</i> – May 2021 |

Skills

- **Languages:** C++, Java, Python, TypeScript, JavaScript, SQL, Rust, Bash/Shell
- **Framework / Tools:** Next.js, Node.js, React, Jenkins, Docker, Selenium, Playwright, CI / CD, Vitest, Git
- **Architecture:** SQL Optimization, Predicate Pushdown, RESTful APIs, Visitor Design Pattern, Semantic Models

Employment

Senior Member of Technical Staff	Oracle	Jul 2024 – Present
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- Engineered cross-system query federation across two distributed codebases, achieving query-result parity between an analytics engine and a transactional backend while preserving multi-table join topology
- Established test infrastructure from zero for a new GenAI SQL product line, authoring 830+ automated tests across 3 codebases and surfacing ~100 pre-release functional and authorization defects
- Accelerated team validation cycles by 83% (60 to 10 mins) by engineering a multi-repository CI/CD pipeline that builds full-stack ephemeral platform instances and automates end-to-end test execution
- Resolved high-severity production incidents for enterprise customers on a 1-in-5 on-call rotation, root-causing query engine defects and backporting C++ patches across releases under full regression validation
- Expanded query engine observability by instrumenting pushdown ratios and connection wait times, resolving counter bugs to restore cache hit rate reporting accuracy
- Built a Rust semantic graph engine, parsing RDF Turtle files to construct entity-relationship graph data structures for interactive schema and ontology visualization

Member of Technical Staff II	Oracle	Jul 2022 – Jul 2024
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- Integrated 3 database targets end-to-end (Vertica, MySQL HeatWave, and a remote engine access via SOAP), configuring driver connectivity, defining pushdown capabilities, and building C++ visitor SQL generators
- Reduced query latency by 75% (40–50s to 7–11s) by designing and building C++ visitor-pattern SQL generators, pushing down complex aggregations and window functions to native columnar engines
- Raised query cache hit rates from 10–20% to 30–45% by replacing exact-match lookups with projection-subset matching, eliminating redundant warehouse round trips across enterprise dashboards
- Resolved a critical customer escalation on Amazon Redshift by implementing temporal predicate pushdown in C++ SQL generators, eliminating query row-limit rejections across cloud pods
- Resolved runtime query failures across Google BigQuery and MySQL dialects, fixing string-literal escaping rules and eliminating a floating-point precision defect in date reconstruction

Member of Technical Staff I	Oracle	Jul 2021 – Jul 2022
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- Implemented cross-module dependency inversion in a fixed-link-order C++ build, using typed callbacks to enable dynamic DSN resolution across isolated query engine components
- Hardened database connectivity by engineering dynamic TLS validation across ODBC C APIs and configurable TCP connect timeouts in DSN generation, replacing multi-minute OS defaults with bounded connection waits
- Eliminated query-cache metadata corruption in C++, tracing silent state failures to omitted copy-constructor fields and uninitialized struct memory during cache insertion

Software Engineering Intern	JPMorgan Chase & Co	Jul 2020 – Aug 2020
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- Engineered an event-driven banking transaction prototype using Apache Kafka for real-time message streaming, earning a full-time software engineering return offer following the internship

Projects

Prizlit - Physical Purchase to Digital Lead Sweepstakes Platform

- Engineered a defense-in-depth authentication layer using Next.js, Redis, and PostgreSQL, implementing atomic Lua sliding-window rate limiting and uniform responses and response-time floors to eliminate account enumeration